Noise Analysis — 10um

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PAGE 1 — FILE STATISTICS

Folder:            data/dataset_hamming_clean/test/10um
Files:             100
Signal length:     2500–2500 samples
Dtype:             float64

Aggregate noise statistics:
  mean(σ):         0.305891
  std(σ):          0.256055
  CV(σ):           0.8371
  mean(RMS):       0.783883
  mean(kurtosis): -0.5173  (0 = Gaussian)

  mean(DC):        0.145054
  std(DC):         0.856918
  Amplitude range: [-2.3559, 3.2959]
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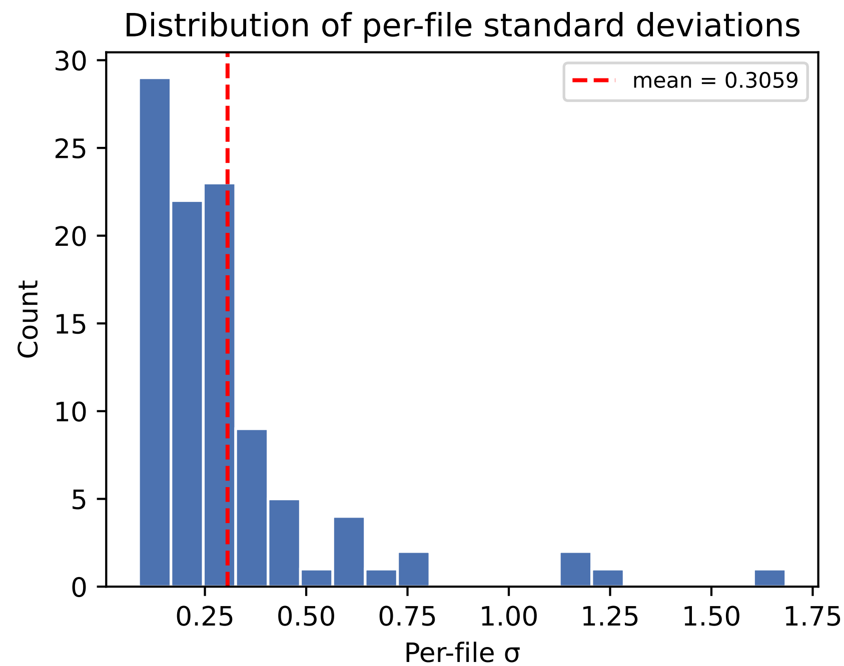
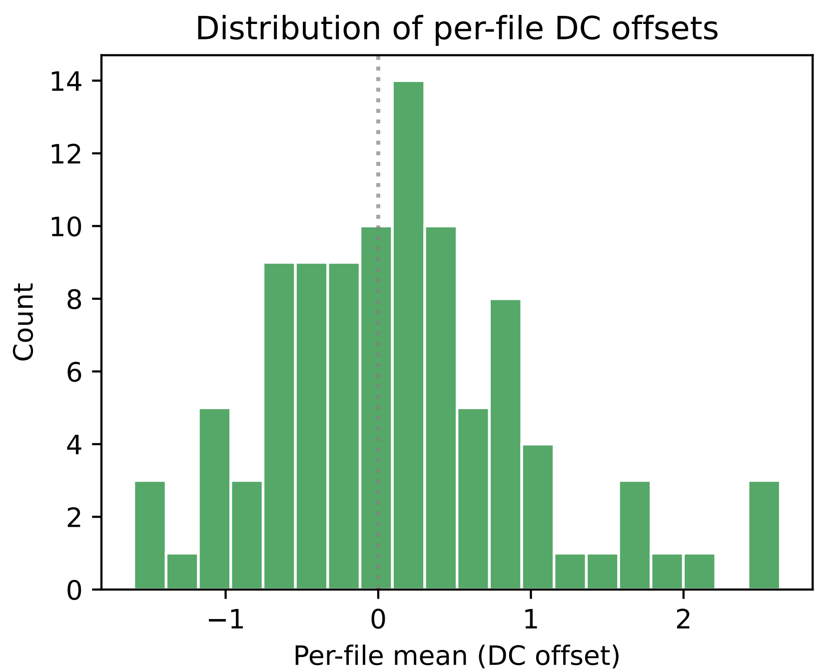
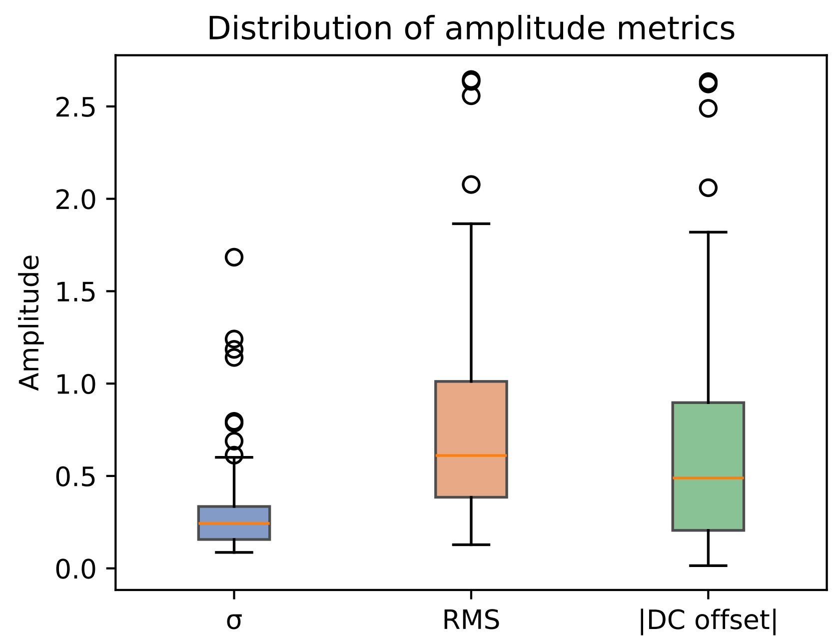
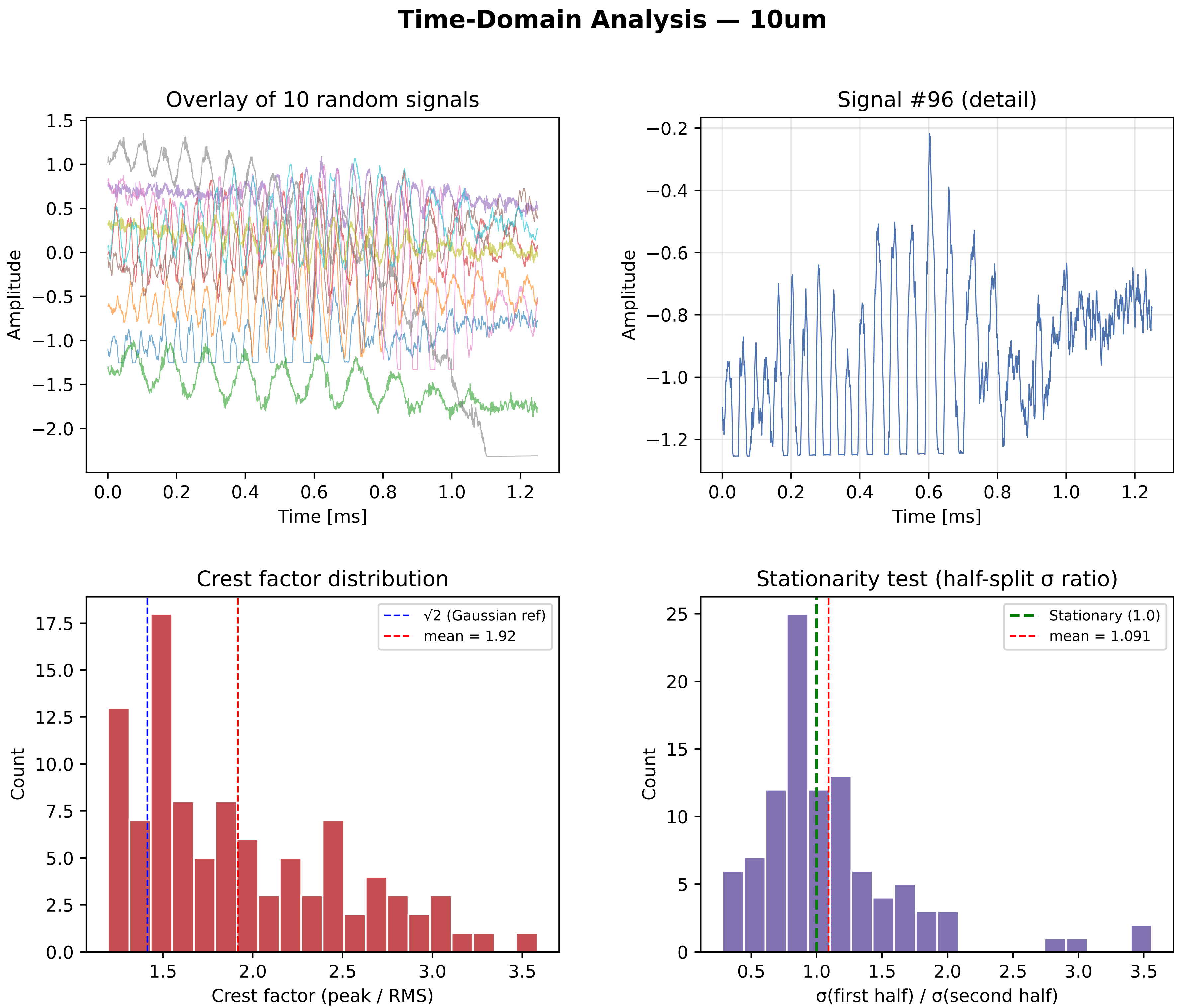


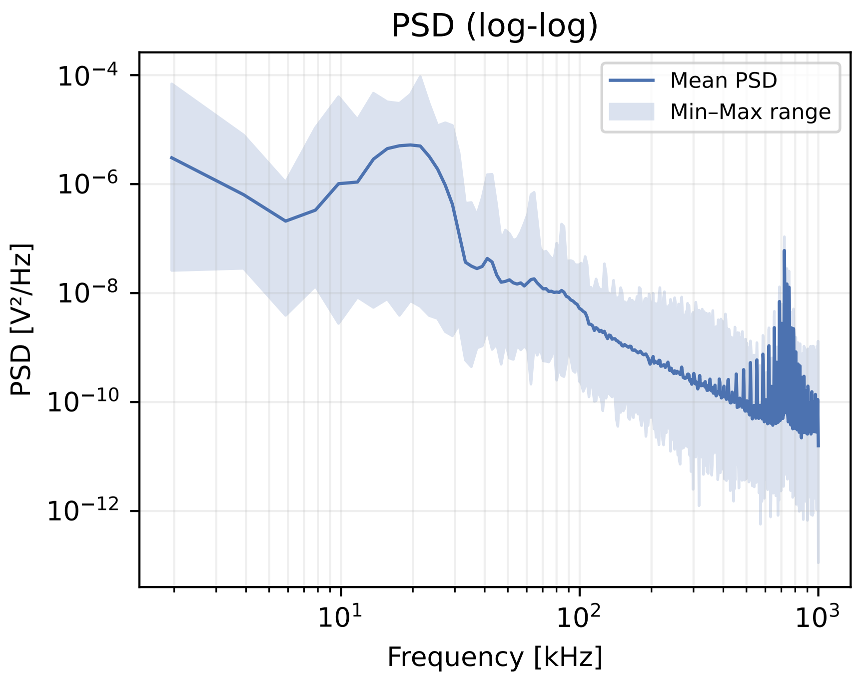


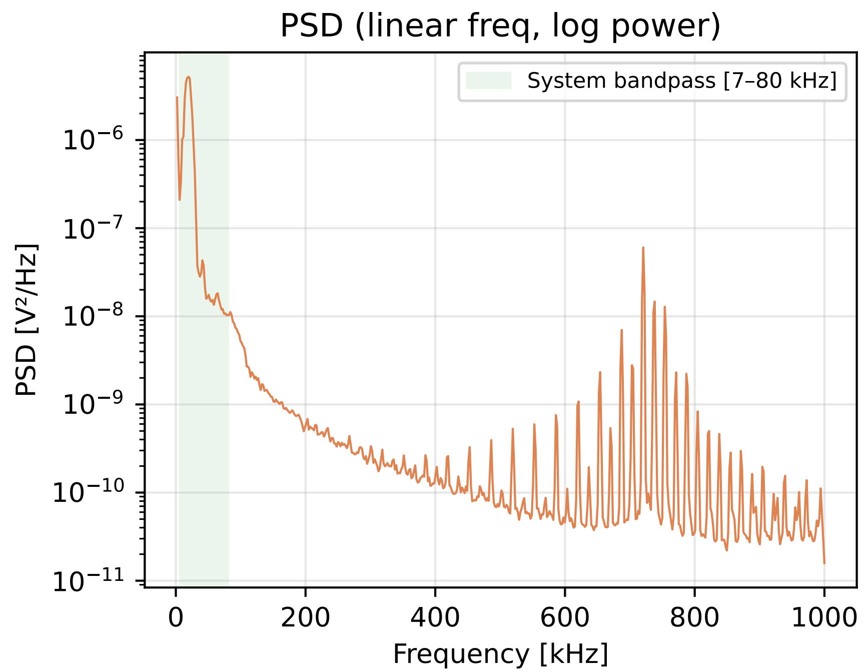


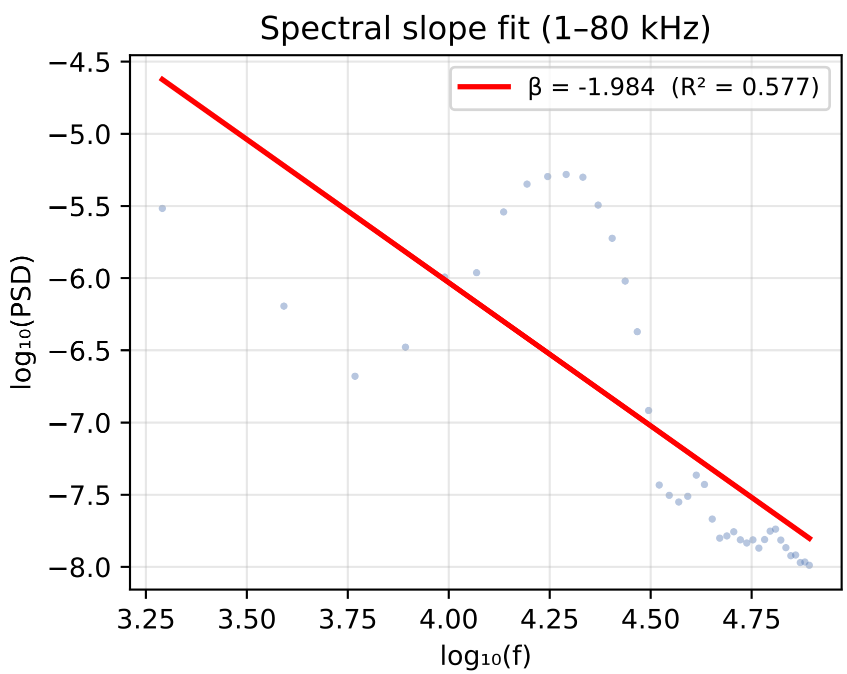


Time-Domain Analysis — 10um
Overlay of 10 random signals
Amplitude
Time [ms]
Signal #96 (detail)
Amplitude
Time [ms]
Crest factor distribution
√2 (Gaussian ref)
mean = 1.92
Count
Crest factor (peak / RMS)
Stationarity test (half-split σ ratio)
Stationary (1.0)
mean = 1.091
Count
σ(first half) / σ(second half)

Power Spectral Density — 10um





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SPECTRAL ANALYSIS SUMMARY

Spectral slope β:      -1.9837
Fit quality R²:        0.5769
Spectral flatness:     0.0040  (1.0 = white)

Slope interpretation:
  β ≈  0 → White noise    (flat PSD)
  β ≈ −1 → Pink / 1/f     (equal energy per octave)
  β ≈ −2 → Brown / 1/f²   (random walk)

Classification:        Brown (1/f²)
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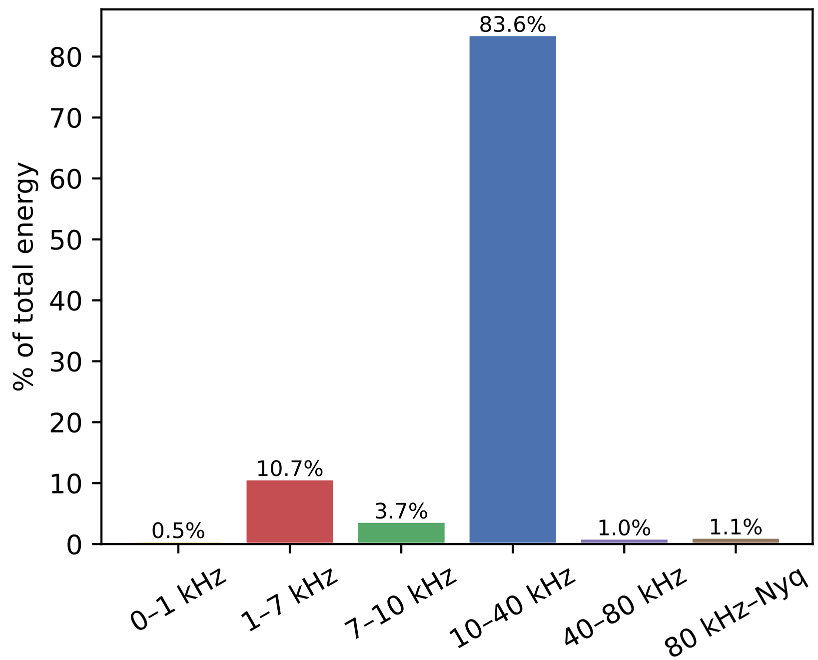
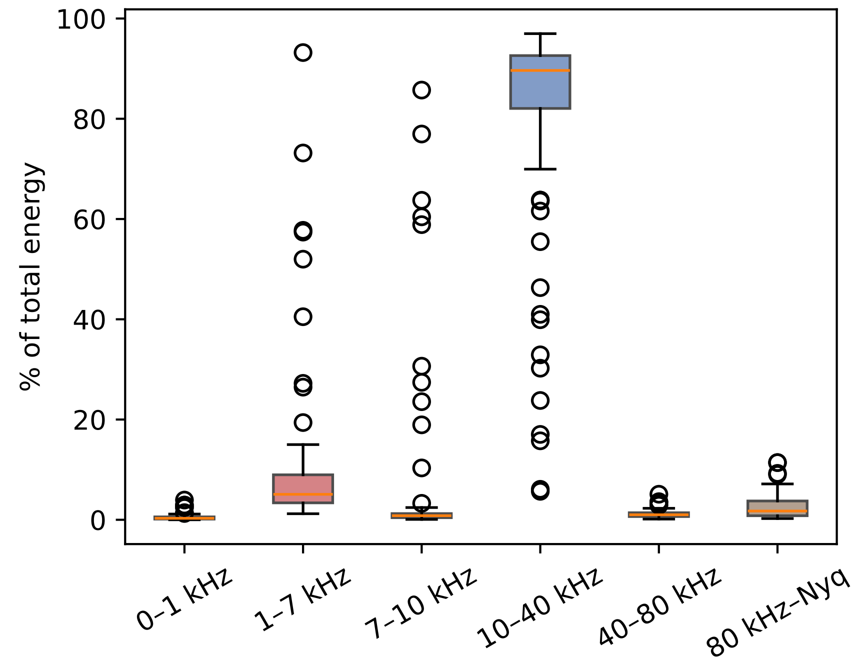
Frequency Band Energy — 10um

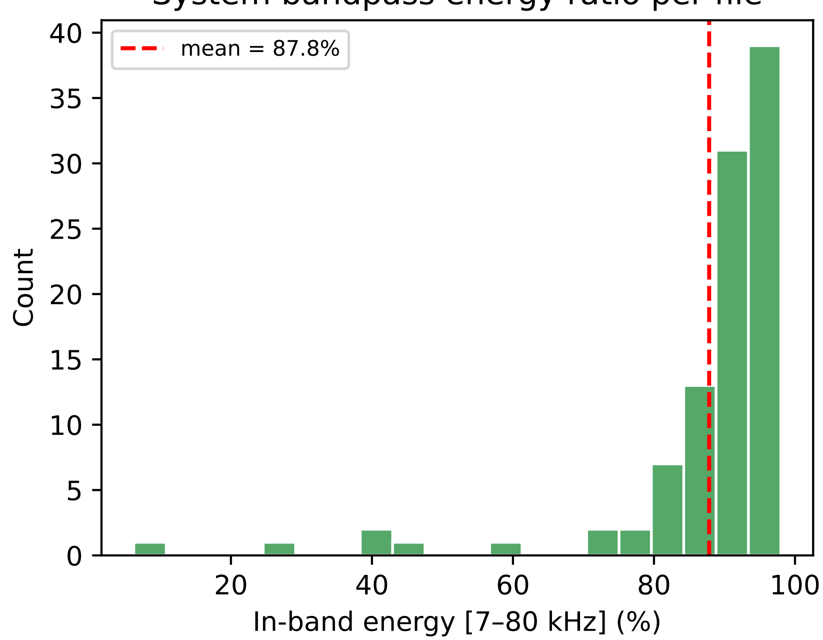
Average energy per frequency band



Per-file band energy distribution



System bandpass energy ratio per file



BAND ENERGY SUMMARY

Band	Mean %	Std %
0–1 kHz	0.5%	0.6%
1–7 kHz	10.7%	14.7%
7–10 kHz	3.7%	15.6%
10–40 kHz	83.6%	20.6%
40–80 kHz	1.0%	0.8%
80 kHz–Nyq	1.1%	2.3%
In-band [7–80k]	87.8%	14.6%

Noise Type Classification — 10um

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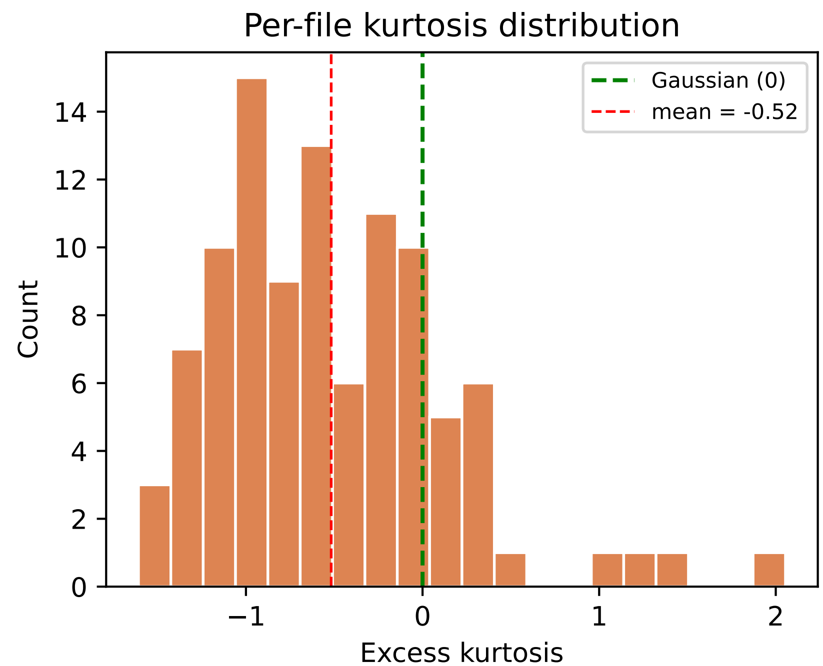
NOISE TYPE CLASSIFICATION

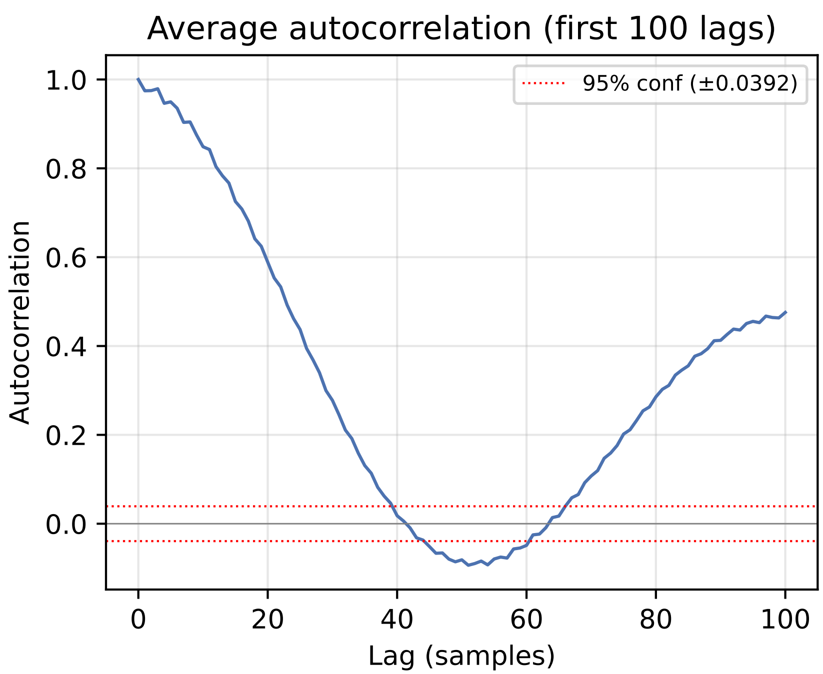
Spectral color:    Brown (1/f²)
  slope β =        -1.9837
  flatness =       0.0040

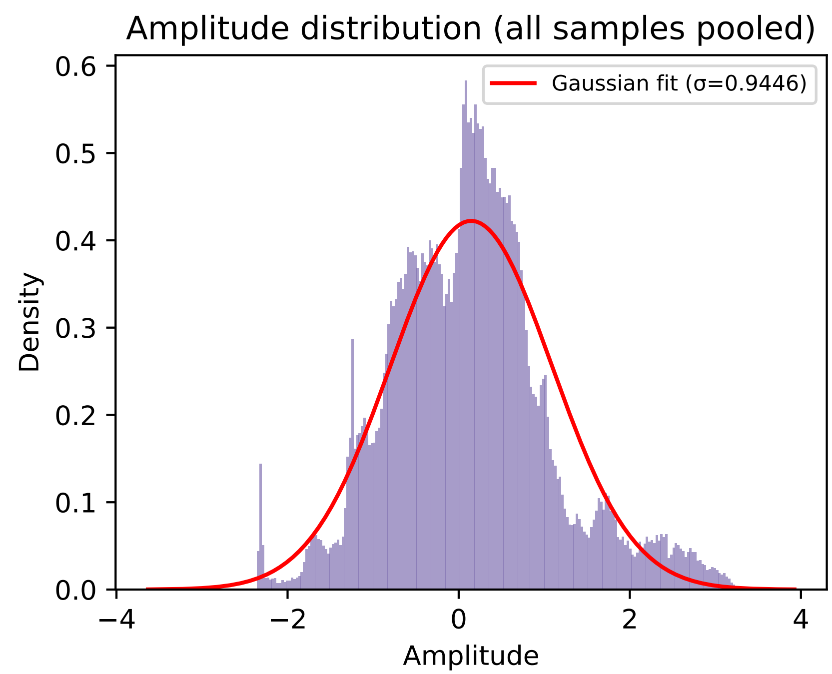
Amplitude dist.:   Near-Gaussian
  kurtosis (mean): -0.5173
  (excess kurtosis: 0 = Gaussian,
   >0 = heavy-tailed, <0 = light-tailed)

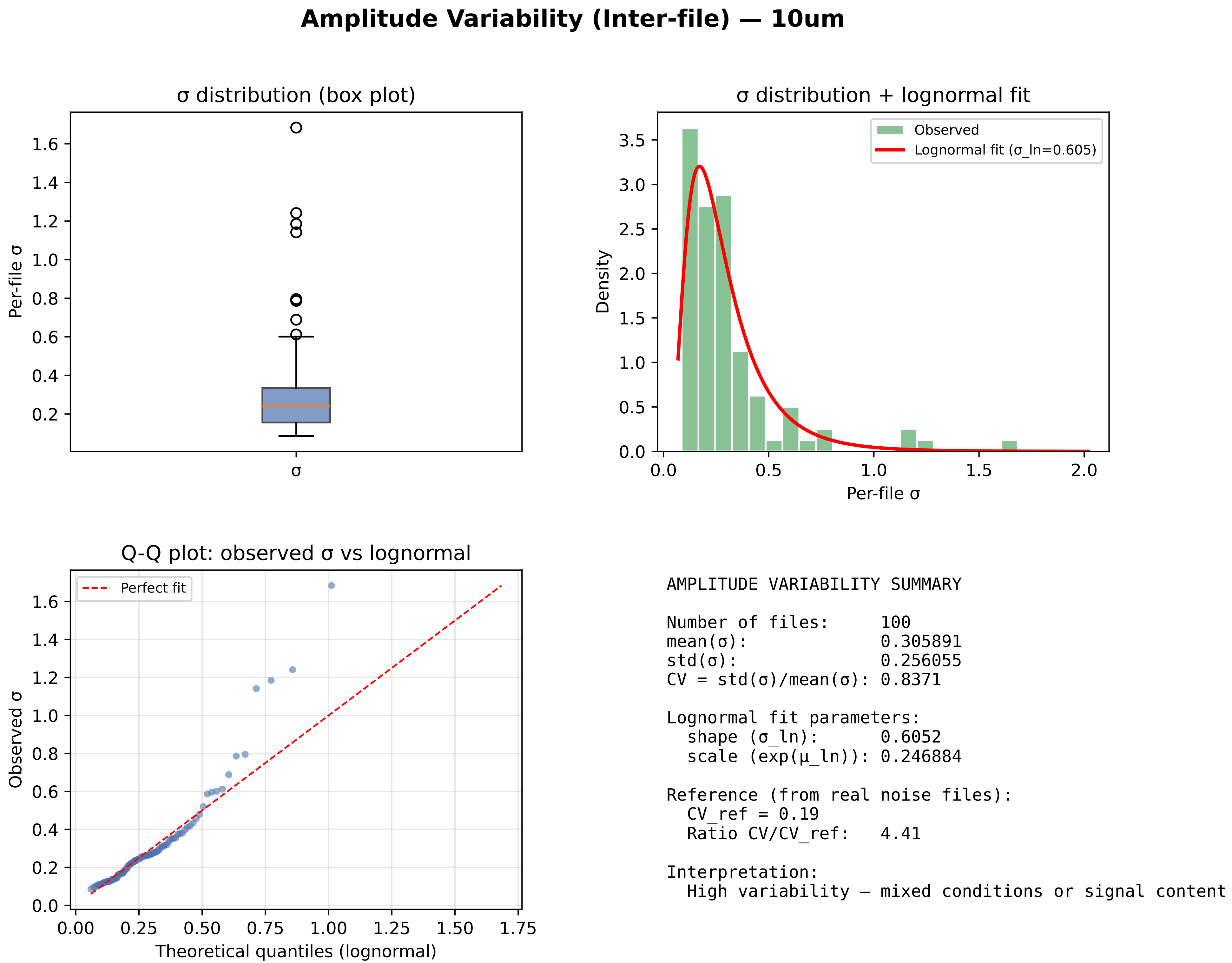
Decision boundaries:
  |β| < 0.3    → White
  0.3–1.5      → Pink (1/f)
  > 1.5        → Brown (1/f²)

  |kurt| < 0.5  → Gaussian
  0.5–2.0       → Near-Gaussian
  > 2.0         → Non-Gaussian
```








Amplitude Variability (Inter-file) — 10um
σ distribution (box plot)
Per-file σ
σ
σ distribution + lognormal fit
Observed
Lognormal fit (σ_ln=0.605)
Density
Per-file σ
Q-Q plot: observed σ vs lognormal
Perfect fit
Observed σ
Theoretical quantiles (lognormal)
AMPLITUDE VARIABILITY SUMMARY
Number of files: 100
mean(σ): 0.305891
std(σ): 0.256055
CV = std(σ)/mean(σ): 0.8371
Lognormal fit parameters:
shape (σ_ln): 0.6052
scale (exp(μ_ln)): 0.246884
Reference (from real noise files):
CV_ref = 0.19
Ratio CV/CV_ref: 4.41
Interpretation:
High variability – mixed conditions or signal content